

Are German executives overpaid? — and if so, how is it hidden?

A study of ~130 German listed firms, 2006–2021. Written so anyone can follow it — no economics background needed.

The big idea in one paragraph

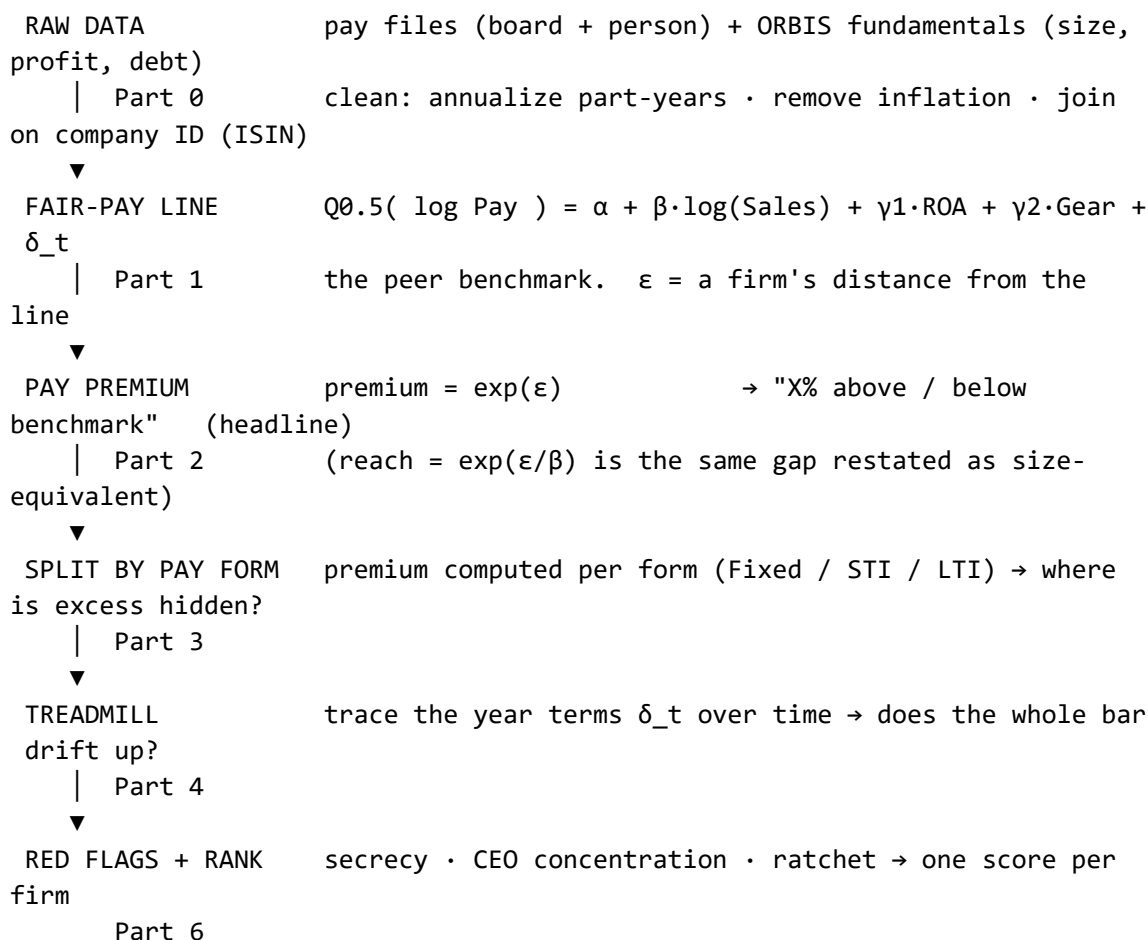
Bigger companies pay their bosses more — that’s normal. So we can’t just say “this CEO earns a lot, therefore overpaid.” Instead we build a **fair-pay line**: *given how big a company is, how profitable, and how indebted, what do similar companies (“peers”) pay their board?* If a firm pays **above** that line, it pays a **premium** over its peers. We turn that gap into one intuitive number — the **pay premium** — then ask three follow-up questions: **where** inside the pay package the excess hides, whether the fair-pay bar **creeps up** over time, and which **governance red flags** light up. (We say “above benchmark,” not “overpaid” — the number is a measured gap, not a verdict.)

How to read every chart in this deck

A few things repeat everywhere — learn them once:

- **premium = 1.0** → paid *exactly* on the peer benchmark. **premium = 2** → **double** the benchmark. **premium < 1** → below it. (*A more dramatic restatement, “reach” = paying like a firm $X\times$ your size, is the same gap in bigger-sounding units.*)
 - **“log” axes** just means each step multiplies (1 → 10 → 100). We use them because firm sizes and pay span huge ranges.
 - **“peers”** = firms of similar size, profitability and debt *in the same year* — not one global average. The fair-pay line **is** the peer comparison (see Step 1).
 - Pay is split into three forms: **Fixed** (base salary + perks), **STI** (Short-Term Incentive = the annual bonus), **LTI** (Long-Term Incentive = shares, options, multi-year bonus).
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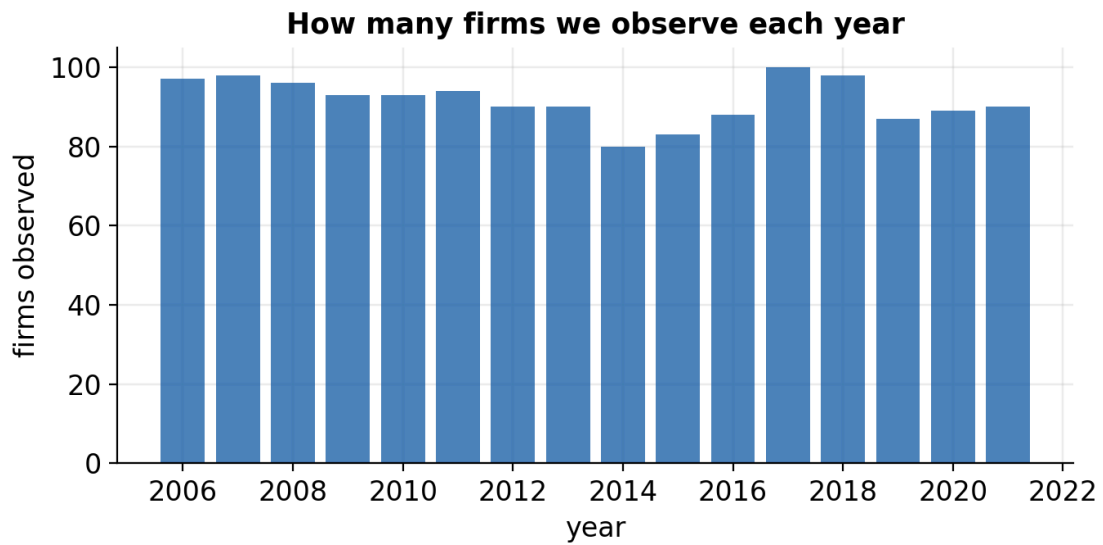
The workflow at a glance



Each step below corresponds to one box. Every arrow feeds the next.

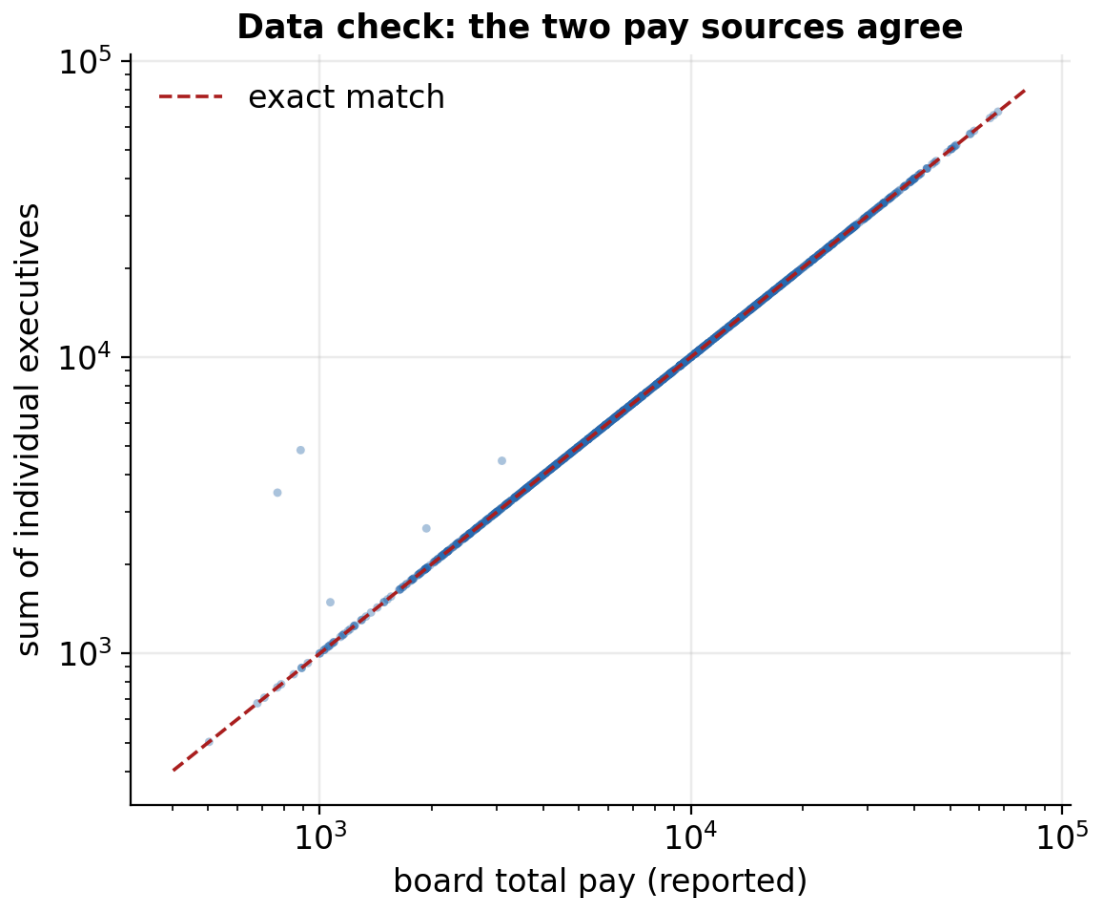
Step 0 — The data

We use three hand-collected pay files (board totals, individual executives, and a person directory) covering **2006–2021**, joined to **ORBIS** company fundamentals (size, profit, debt). After cleaning — annualizing partial years, removing inflation, and de-duplicating ORBIS — we have **~130 non-financial German firms** with everything needed (banks, insurers and foreign-listed firms are excluded because the size measure doesn't apply to them).



Panel coverage

How to read it: each bar = how many firms we can analyze in that year. **What it tells us:** coverage is stable across the whole 2006–2021 window (~70–90 firms/year), so time comparisons are fair — we’re not comparing a handful of firms early on to many later.



Reconciliation

How to read it: each dot is one company-year. Horizontal = the sum of the individual executives' pay; vertical = the separately reported board total. The dashed line is “they match exactly.” **What it tells us:** the dots sit on the line (1,265 of 1,270 within 5%) — the two data sources agree, so the cleaning is trustworthy.

Step 1 — What is “normal pay”? The peer benchmark

This is the heart of the study. We fit one **fair-pay line** across all firms, at the **median** (the typical firm, so a few giant packages don't drag the bar up):

$$Q_{0.5}(\log \text{total_comp}_{it}) = \alpha + \beta \log(\text{opre}_{it}) + \gamma_1 \text{roa}_{it} + \gamma_2 \text{gear}_{it} + \delta_t$$

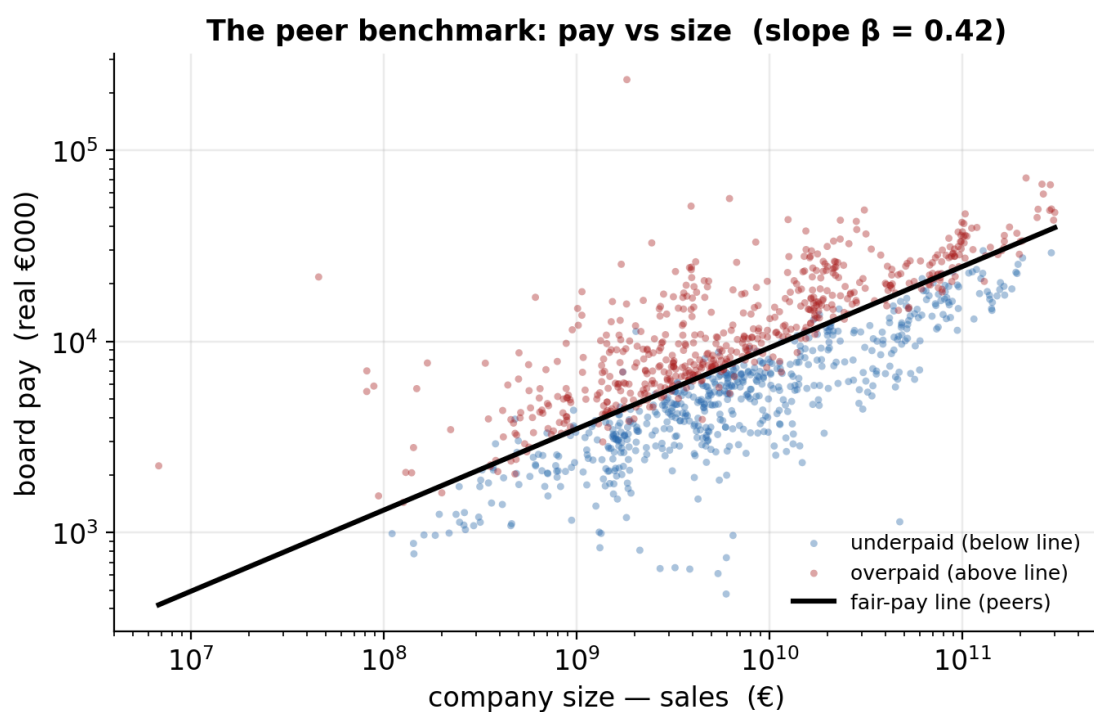
What $Q_{0.5}(\cdot)$ means: it is *not* a transformation of the data — the dependent variable is just **log total_comp** (one number per firm-year). $Q_{0.5}$ says we fit the line through the **conditional median** (half pay more, half less) instead of the av-

erage. Fitting the median — minimizing absolute, not squared, errors — is what keeps a few €20M packages from dragging the line up. Same variable, sturdier line.

Median over which set? Over all **1,161 firm-years at once**, but *conditionally*: one line minimizing total $|\text{pay} - \text{line}|$, so that at each size/profit/debt/year ~half the firms sit above it and half below. Not a global median (that would ignore size), not separate medians per group — a single line threaded through the middle of the cloud at every size.

In plain words: *normal (log) pay = a baseline + $\beta \times (\log \text{ company size})$ + a profitability adjustment + a debt adjustment + a “what year is it” adjustment.* opre = sales (size), roa = profitability, gear = debt. The model predicts the **typical peer’s pay** for any combination of these.

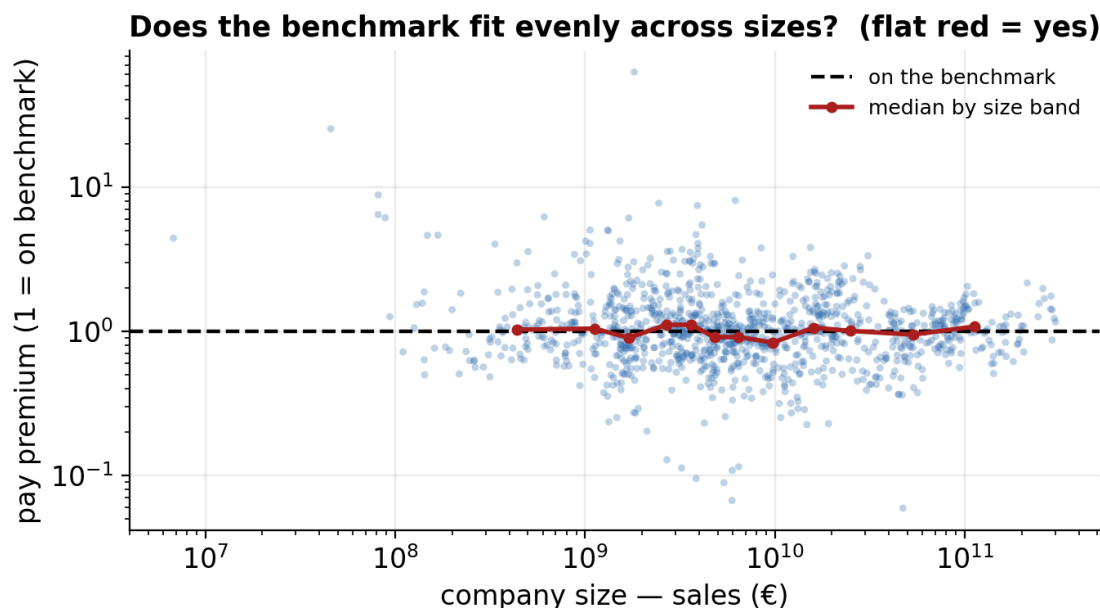
Why this is a peer comparison: two firms with the same size, profit, debt and year get the **same** predicted pay — they are peers. A firm’s gap from this line is how far above its peers it is paid. We do *not* lump everyone into one average; the line bends with each firm’s characteristics.



Benchmark fit

How to read it: each dot is a company-year — horizontal = company size (sales), vertical = board pay, both on log axes. The solid line is the **fair-pay line** (normal pay for that size). **Red dots sit above the line (overpaid); blue below (underpaid).** What it tells

us: pay clearly rises with size (the line slopes up, $\beta \approx 0.42$: every doubling of size adds ~34% pay), and firms scatter above and below — those distances are exactly what the **pay premium** measures next.



Fit across firm size

How to read it: each dot is a company-year — horizontal = size, vertical = its **pay premium** (actual ÷ benchmark; 1 = on the line). The red line is the median premium within each size band. **What it tells us:** the red line stays **flat at 1× across all sizes** — the benchmark fits small and large firms equally well, with no size bias or curvature. So the leftover gaps are real signal, not an artifact of the size measure. (*Plotting the gap against size is the honest fit check: a good benchmark leaves no pattern across size.*)

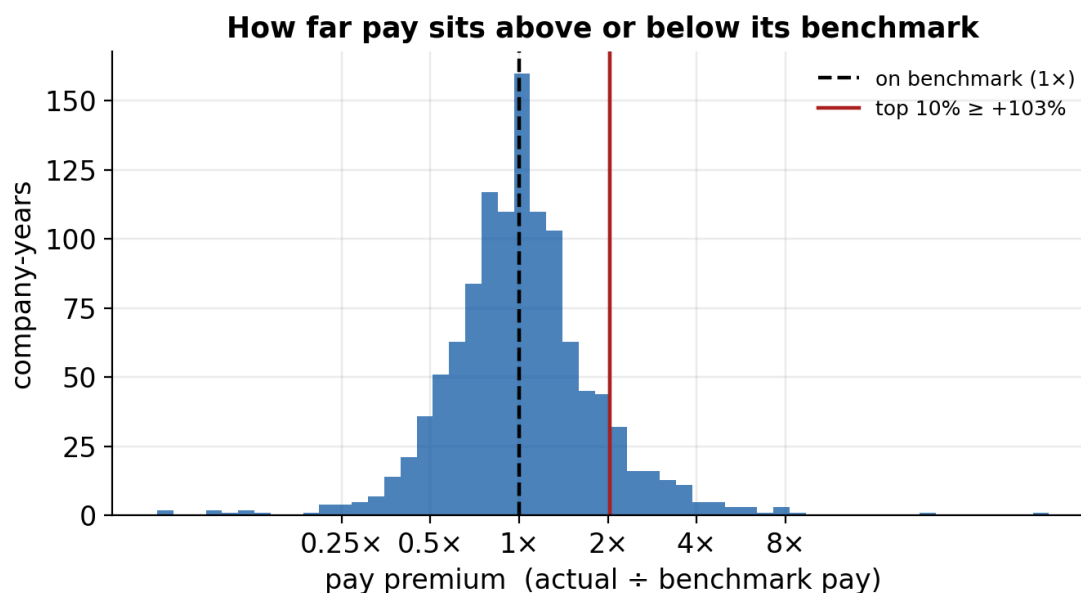
Step 2 — How far above the benchmark? “Pay premium”

A gap of “+0.7 in log-pay” means nothing to most people. So we turn the distance ε into one honest number:

$$\text{pay premium}_{it} = \exp(\varepsilon_{it}), \quad \varepsilon = \text{distance from the benchmark line (in logs)}$$

In plain words: premium = 1.2 → pay is **20% above** benchmark; premium = 2 → **double**; premium = 0.8 → 20% **below**. We say “above benchmark,” **not** “overpaid” — it’s a measured gap from peers, not a verdict.

(A vivid second view, used in a couple of charts, is the **size-equivalent reach** $= \exp(\epsilon/\beta)$. Because the size-pay slope is gentle ($\beta \approx 0.4$), a $2\times$ premium corresponds to “paying like a firm $\sim 5\times$ your size.” Same information, more dramatic units — premium stays the headline.)

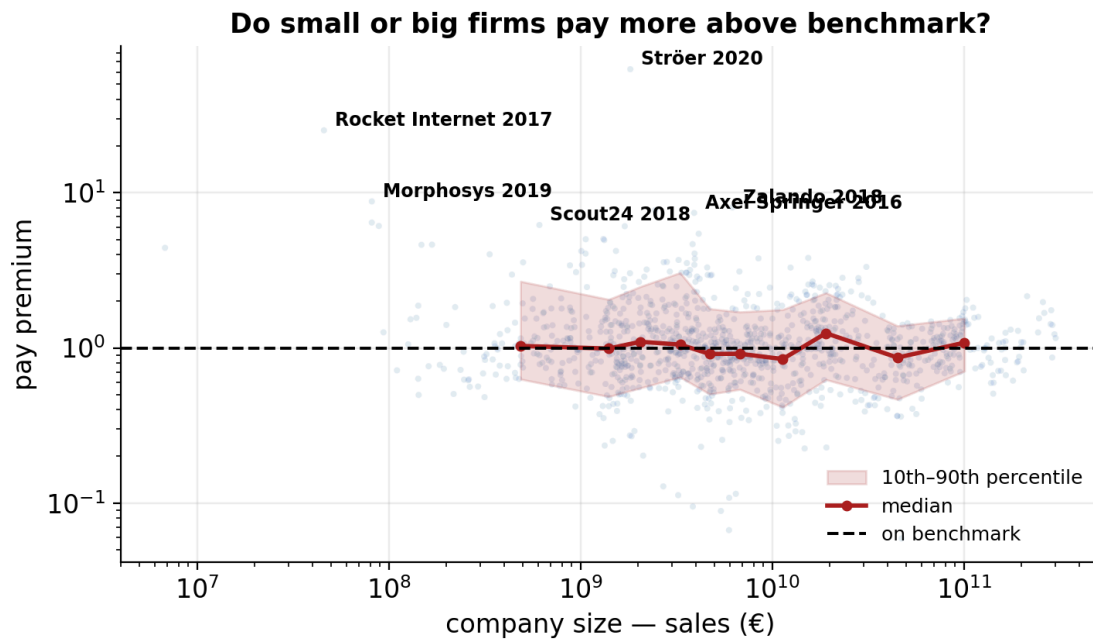


Pay-premium distribution

How to read it: each bar counts company-years at a given premium (actual ÷ benchmark pay). The dashed line is “on benchmark” (1×); the red line marks the **top 10%** threshold.

What it tells us: the curve is **centred at 1× by design**, but it is **wide** — the **top 10% pay ≥ 2× their benchmark** (+103%), with a thin tail far beyond. The story is the *spread*, not the centre.

⚠ **Why this is a bell curve (important).** The benchmark is fitted *to these same firms* at the median, so the typical firm is “on benchmark” automatically. This chart therefore **cannot** say “the market overpays” — only **who** sits in the expensive tail. Whether the whole bar drifts upward over time is a *separate* question — that’s Step 4.

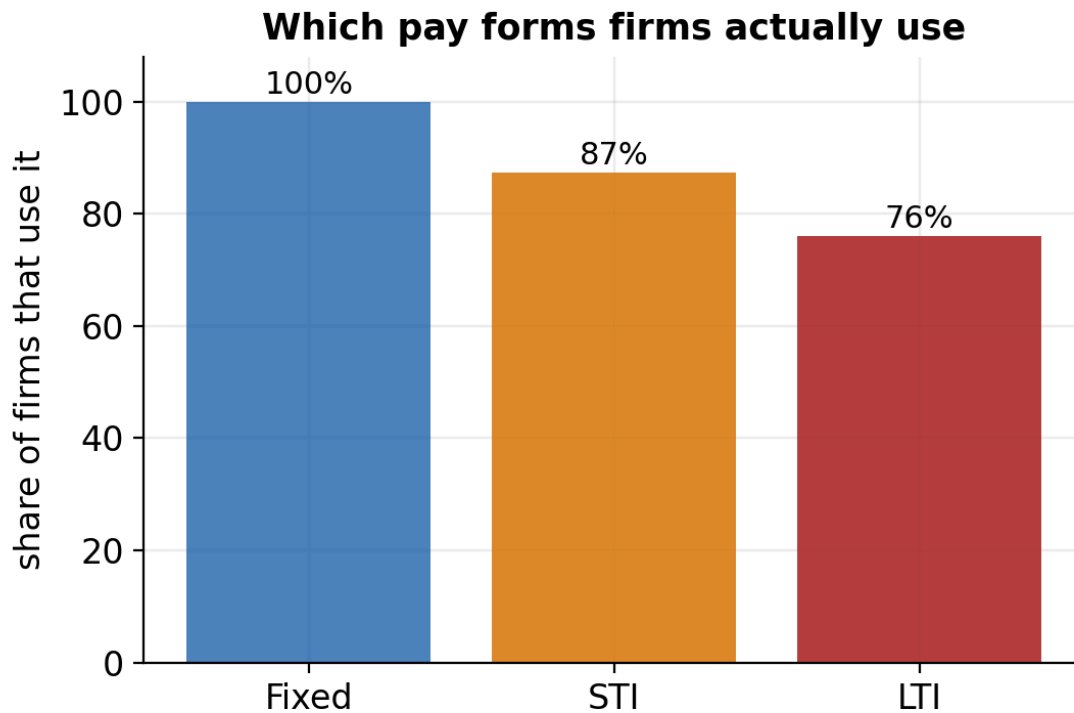


Premium vs size

How to read it: horizontal = firm size; vertical = pay premium. The red line is the **median** by size; the shaded band is the **10th–90th percentile** (the full spread, small firms and big firms); the most extreme firms (premium > 4×) are **named**. **What it tells us:** the median sits near 1× at **every** size (no small- vs big-firm tilt), but the **10–90 band is wide everywhere** — within any size group some pay far above benchmark, some below. The named outliers (Ströer, Rocket Internet, Morphosys, Porsche Holding ...) are mostly **small firms or holding companies** with little operating revenue but a full board to pay — worth knowing they drive the extreme tail.

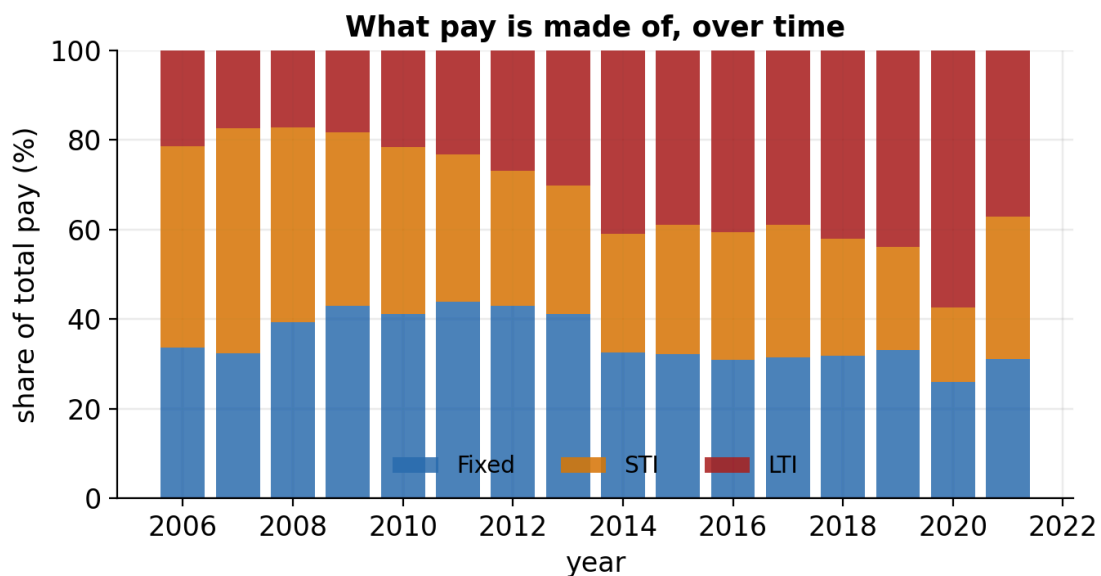
Step 3 — Where is the overpayment hidden?

Total pay can look reasonable while the excess is piled into one component. We benchmark **each pay form separately** (Fixed / STI / LTI) against its own size curve, then build the **headline as the parts combined** — the overall premium is literally (*sum of the forms*) ÷ (*what the forms should be*), so it's internally consistent. The question: **is any single form stretched far above the firm's own average?**



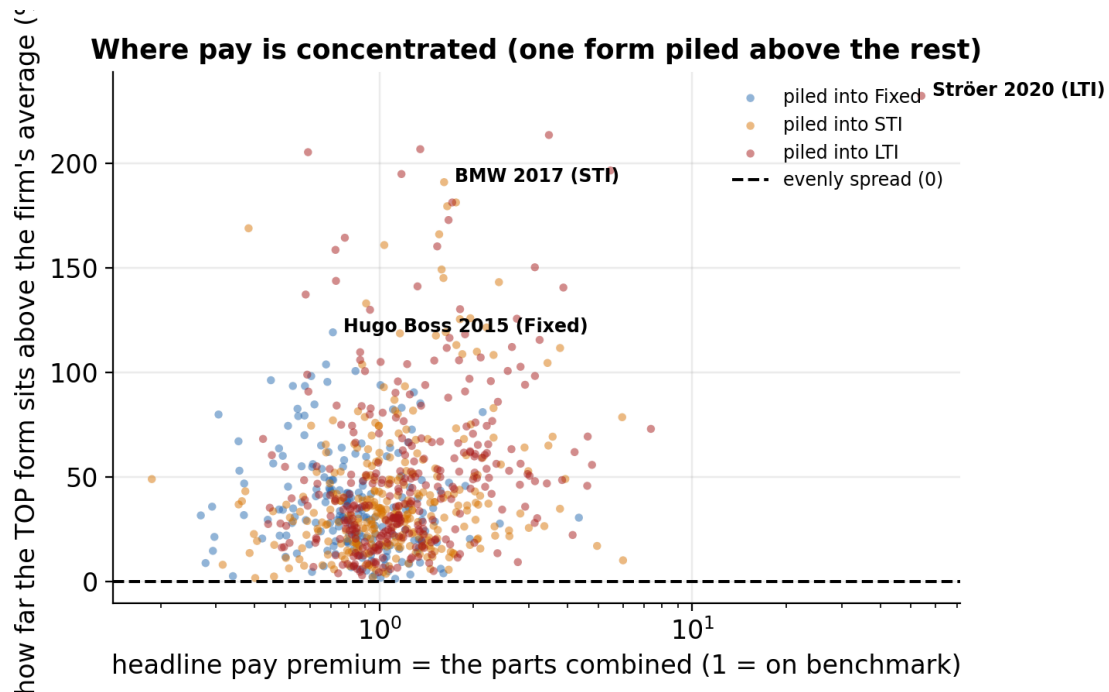
Bucket coverage

How to read it: each bar = the share of firms that actually use that pay form. **What it tells us:** almost everyone pays **Fixed** salary (~100%) and an annual bonus (**STI**, ~87%), but only ~76% grant **LTI** (long-term/equity). That unevenness is why we treat the forms separately rather than lumping them.



Pay mix over time

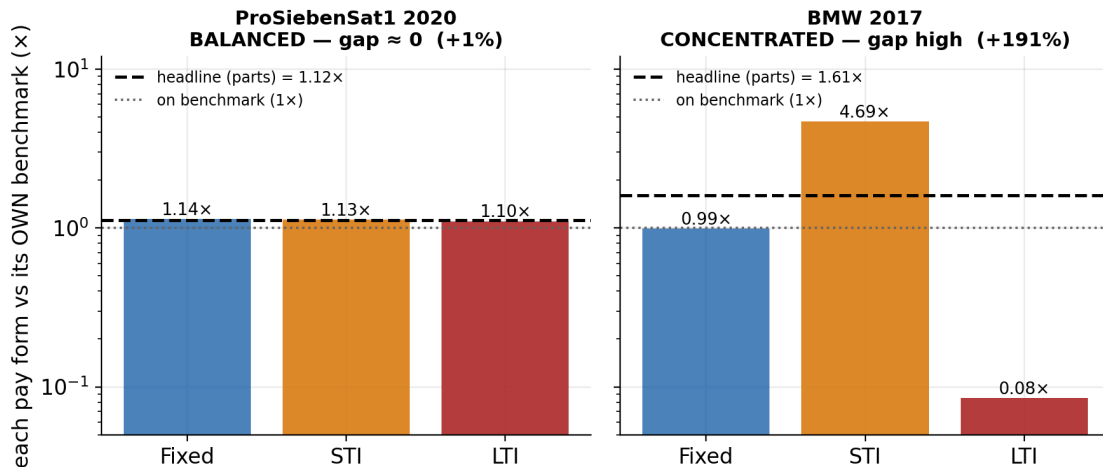
How to read it: each bar is a year; the colored segments show what share of total pay was Fixed / STI / LTI. **What it tells us:** the structure shifts visibly — **LTI (long-term/equity, red) roughly doubles after ~2013**, from ~20% to ~40% of pay, mostly at the expense of the annual bonus (STI). More money moves into the form that’s hardest for outsiders to scrutinize — exactly where Step 3 finds the worst concealment.



Where pay is concentrated

How to read it: each dot is a company-year. Horizontal = the **headline premium** (the parts combined); vertical = **how far the single most-stretched form sits above the firm's own average (%)**. The dashed line at 0 = pay spread evenly; colour = which form is piled up; the biggest case in each form is **named**. **What it tells us:** the median firm's top form sits **+35% above its own average**, the worst cases **+300% and beyond**. The big concentrations are overwhelmingly in **LTI / equity** (red) and **bonuses** (STI, orange) — e.g. **Morphosys** piles it into the annual bonus, **Ströer** and **SAP** into equity. That pile-up is the “hidden stretch.”

'Hidden stretch' = how far the top form sits above the firm's own average



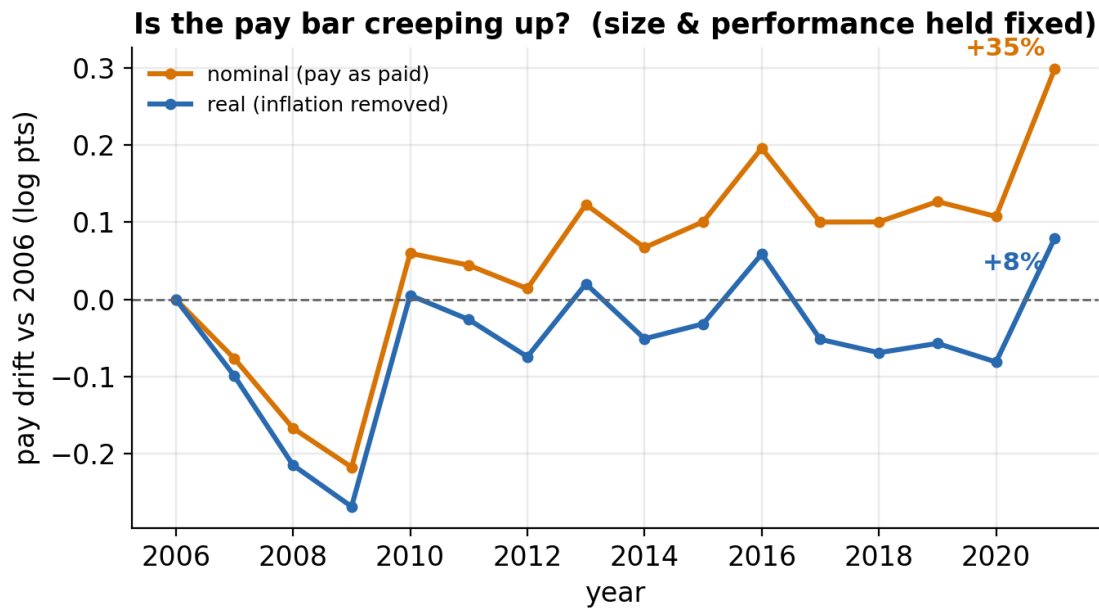
Concentration explained

Worked example. The headline here is the sum of the parts: each form is measured against its own benchmark, and the overall premium is simply what those forms add up to. (This matters — a *separate* figure for “total pay” can disagree with the parts, because the reported `total_comp` mixes in a *realized* multi-year bonus that the grant-based parts leave out; for Gerresheimer 2017 the parts summed to €4.06M while reported `total_comp` was €5.78M.) Because the headline *is* the parts combined, the most-stretched form is always at or above it, so the gap reads cleanly as “*how far the top form pokes above the firm’s own average.*” Each bar is one form vs its own benchmark; the dashed line is the headline:

- **Left (balanced):** all three forms sit near the headline $\rightarrow$ gap ≈ 0 , pay spread evenly.
- **BMW 2017 (concentrated):** the bonus (STI) is 4.7 $\times$ its benchmark while salary is normal and equity near zero $\rightarrow$ the bonus pokes far above the headline $\rightarrow$ big gap. The excess is **piled into the annual bonus**.

Step 4 — Is the fair-pay bar creeping up? (the treadmill)

Even if no single firm looks scandalous, the **bar itself** can drift up: same size, same performance, more pay every year. The “what year is it” adjustments δ_t from Step 1 trace this directly.

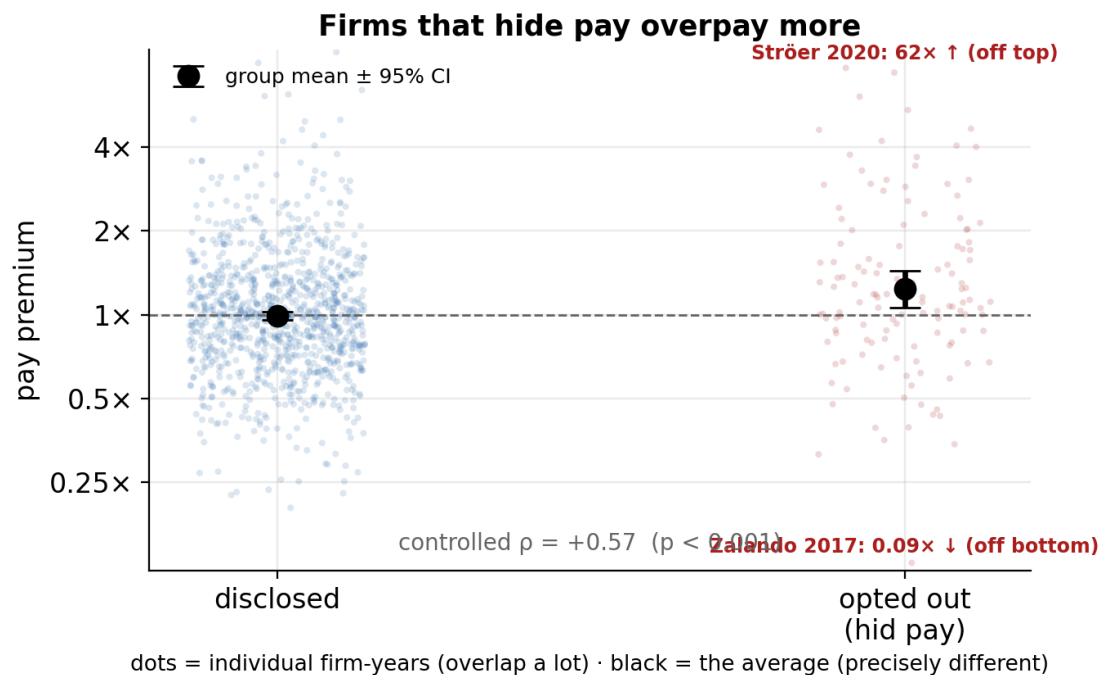


Treadmill

How to read it: horizontal = year; vertical = how much the *benchmark* shifted, holding size and performance fixed (0 = 2006). **Orange = nominal** (euros as paid); **blue = real** (inflation removed). A rising line = a treadmill. **What it tells us:** the bar **does creep up** — **but mostly through inflation**. Nominal benchmark pay ends +35% above 2006; once inflation is stripped out only +8% is left. So there is a treadmill in headline euros, yet the genuine, inflation-free drift is small. (Both lines dip through the 2008–09 crisis.)

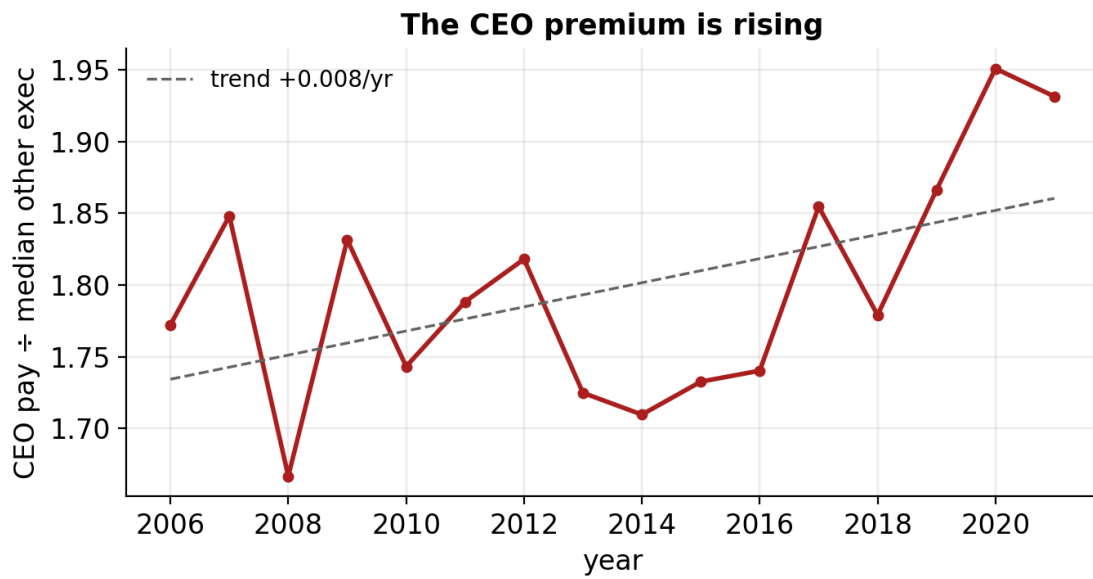
Step 5 — Governance red flags

Three independent warning signs, each a simple test.



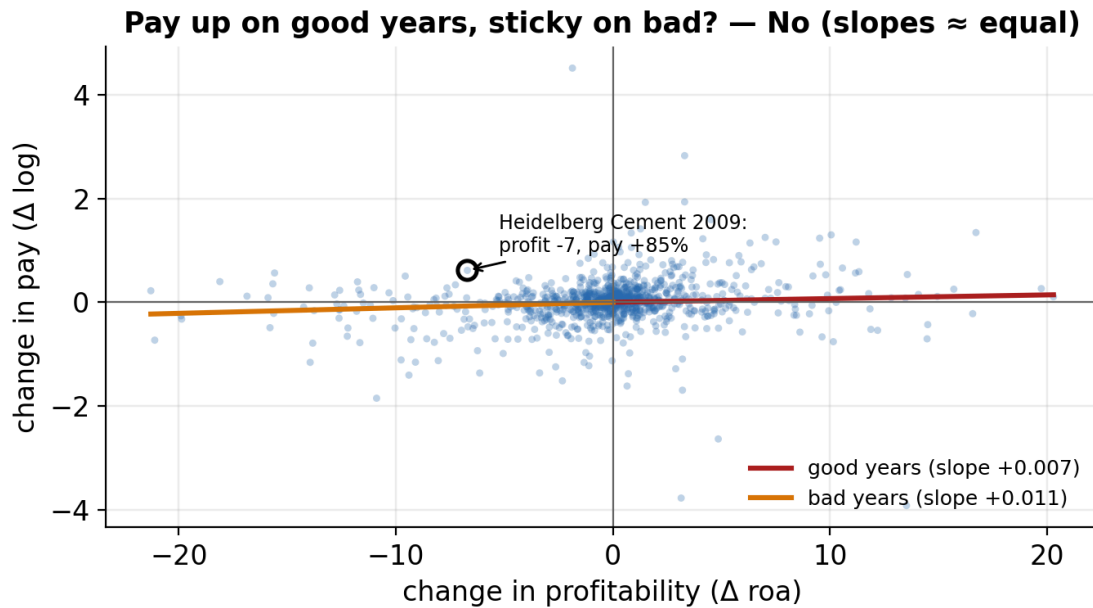
Secrecy

How to read it: two groups — firms that **disclosed** individual pay vs. those that **opted out**. Faint dots are individual firm-years; the **black marker is each group's average**, and the small vertical bar is its **95% confidence interval** — how precisely we know that average. **What it tells us:** the **individual dots overlap heavily** — firms vary a lot either way, so a raw side-by-side looks inconclusive. But the **averages are precisely different**: the opted-out average sits clearly higher and its confidence interval doesn't even touch the disclosed one. Controlling for size/profit/debt/year, opting out goes with **about +77% higher pay** — a statistically strong result, very unlikely to be a fluke ($p = +0.57$, $p < 0.001$). The lesson: *a wide spread doesn't mean there's no signal* — a real difference in **averages** survives even when individual firms overlap.



CEO concentration

How to read it: the line is, each year, how many times more the CEO is paid than the *median other* board member. **What it tells us:** the CEO premium **rises over time**, from $\sim 1.8\times$ to $\sim 1.9\times$ — pay is becoming more concentrated at the very top.

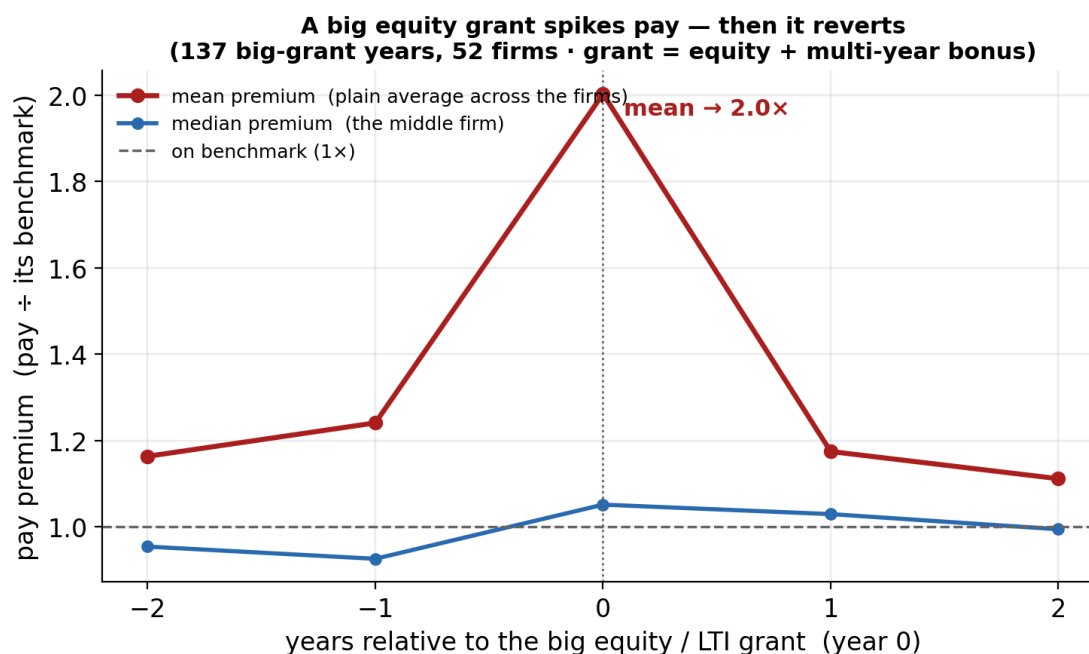


Asymmetric ratchet

How to read it: each dot is a company-year change — horizontal = change in profitability, vertical = change in pay. Separate lines fit the good years (right) and bad years (left). **What it tells us:** the two slopes are **almost identical** (both near zero) — in this sample pay does **not** ratchet up on good news while staying sticky on bad. This red flag simply does **not** fire — a genuine no-effect result, reported honestly.

Bonus — does a big equity grant inflate the pay premium?

First, what counts as a “grant” here: the **equity / long-term-incentive (LTI) award** — shares + options + the multi-year bonus *granted* that year. It is **not** the annual cash bonus (STI), and **not** a salary increase. A **big-grant year** is one where that LTI award is **more than double the firm’s own usual level** (and worth >25% of pay). We then watch the premium in the years around it.



Grant spike

How to read it: year 0 is the big-grant year. At each year we take all the flagged firms and average their premium two ways — the **mean** (the plain average across those firm-years; at year 0 that’s the average of all 137 big-grant firm-years) and the **median** (the middle firm). The premium itself is each firm-year’s pay ÷ its benchmark. **What it tells us:** the answer is **yes — but temporarily**. In the grant year the **average premium doubles to ~2.0×**, then **reverts the very next year to ~1.1×**. The **median barely moves** (0.95 → 1.05), and that gap between mean and median *is* the story: the spike is a **tail**

phenomenon, driven by a few firms (Zalando, Ströer, Rocket Internet) handing out *enormous* one-off grants that send their pay to multiples of the benchmark for a single year, while the typical grant year is modest.

Why it matters: much of the “expensive tail” is **transient grant timing, not permanent over-payment**. (Part of the spike is mechanical — the grant *is* part of total pay — but the clean doubling-and-reversion is the point.) It separates two very different firms: a *persistently* expensive one (**Hella**, median 2×) from one that merely had a *one-off mega-grant year* (**Zalando 2018**). For governance, the persistent one is the real concern.

The deliverable — which firms to look at first

Every firm gets a score combining the four per-firm flags (above-benchmark pay, hidden stretch, secrecy, CEO concentration), each as a rank among peers. The most-flagged firms:

Rank	Company	Premium	Hidden	Secrecy	CEO ratio	Score
1	Hella	2.0×	+161%	80%	4.0	0.96
2	Varta	1.3×	+119%	50%	22.4	0.90
3	Axel Springer	3.4×	+49%	100%	—	0.89
4	Puma	1.8×	+51%	77%	6.8	0.88
5	CompuGroup	2.1×	+77%	0%	4.4	0.80
6	Rhön-Klinikum	1.1×	—	9%	3.0	0.79
7	Heidelberg Cement	2.0×	+40%	31%	2.3	0.78
8	CTS Eventim	1.3×	+83%	0%	3.7	0.75
9	Sky Dtl.	2.9×	+48%	0%	3.7	0.75
10	ProSiebenSat1	1.6×	+45%	6%	2.2	0.75
11	Hugo Boss	1.2×	+68%	56%	1.8	0.73
12	Stada	1.3×	+64%	0%	2.3	0.69
13	SAP	2.4×	+48%	0%	2.1	0.68
14	Morphosys	6.3×	+62%	0%	1.8	0.67
15	Bertrandt	0.9×	—	100%	—	0.66

Reading the table: **Premium** = the firm’s median pay ÷ its benchmark (1× = on benchmark); **Hidden** = median concentration — how much more the top pay form gets than the firm’s own average; **Secrecy** = share of years the firm hid individual pay; **CEO ratio** = CEO ÷ median other executive (a “—” means the firm opted out, so individual pay is unknown).

What we found

1. **By design the median firm is on benchmark; the story is the spread** — the **top 10% pay $\geq 2\times$ (+103%) their benchmark**, and this expensive tail appears across all firm sizes (not just the giants).
2. **Pay is rarely spread evenly across forms** — the median firm piles **+35% more** into its top pay form than its own average, and the biggest concentrations are in **long-term/equity (LTI)** and **bonuses (STI)**.
3. **The treadmill is weak in real terms** ($\sim +8\%$ over 15 years) — German exec pay did not run away once inflation and firm growth are removed.
4. **Secrecy predicts higher pay** — opting out of disclosure goes with $\sim 77\%$ higher pay (premium), controlling for size/profit/debt/year.
5. **The CEO premium is rising** ($\sim 1.8\times \rightarrow 1.9\times$).

What we deliberately did *not* do (caveats)

- **No individual-mover analysis (Part 5)** — only 70 executives ever change firms (4.8%), too few for that method.
- **Peers are continuous, not industry groups** — we compare to firms of similar size/profit/debt, not to a named industry peer set (a reasonable alternative, not done here).
- **The inflation index is a placeholder** — swap in official Destatis CPI for final numbers (it doesn't affect the premium, only the treadmill's exact size).
- **Lumpy one-off grants make some firms swing year to year** — e.g. **Zalando's premium is $\sim 8\times$ in grant years (2013, 2018) but $\sim 0.1\times$ in others**, so a chart that labels a firm shows the *specific year*; we report medians/percentiles (robust to these spikes) rather than means.